Mahdi Bachraoui

Freelance Mechanical Design Engineer — SolidWorks (CSWP) · 2D Drawings · GD&T / ISO GPS

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Portfolio: **mahdibachraoui.github.io** | LinkedIn: **linkedin.com/in/mahdibachraoui**

I turn sketches, old drawings and rough models into SolidWorks parts and release-ready 2D drawings that a workshop can quote and make. Certified SolidWorks Professional, with 12 months releasing AUDI harness drawings at a Tier-1 supplier under formal change control. Remote projects in English, French or Arabic.

CSWP Certified SolidWorks Professional	**−73%** mass on a motor bracket, redesigned from block to bent sheet	**100+** engineering changes released on AUDI harness drawings	**24 h** fixed quote after you send your files

WHAT I DELIVER

3D modelling	SolidWorks parts, assemblies and sheet metal, from sketches, old drawings or existing models
2D drawings	Production drawings to ISO standards: sections, bills of materials, flat patterns and bend tables
GD&T	ISO GPS tolerancing, drawing checks and tolerance stack-ups (worst case, RSS, Monte Carlo)
Design review	Manufacturability review, DFMEA (AIAG-VDA) and material selection
Documentation	Technical documentation and design calculation sheets
Simulation	Stress and deflection checks on parts and brackets in SolidWorks Simulation
File formats	SolidWorks, STEP, DXF and PDF

DESIGN STUDIES (PORTFOLIO)

Self-initiated studies, drawn and calculated but not built; all sheets are on the portfolio site.

Motor support · sheet-metal redesign — 2026

- Converted an unmanufacturable 16.6 mm block into 5 mm EN AW-5754 sheet: laser cut, three bends, flat pattern, bend table
- Mass down from about 3.07 kg to 0.84 kg (−73%), with a marine-grade alloy in place of pure aluminium

Vertical axis redesign and DFMEA · marine cleaning system — 2026

- Found through an AIAG-VDA DFMEA that a rotating 8.5 m screw would run far above its first critical speed; redesigned with a fixed screw and belt-driven nut
- Moved Action Priority from 4 High and 2 Medium to none at design stage, with 7 actions held open pending physical test

Bearing preload tolerance stack-up and rotating nut · worst case, RSS, Monte Carlo — 2026

- Showed general tolerances put only 4.3% of bearing preloads in window; specified a measured shim and drew the nut to suit

Nozzle carriage · redesign in marine-grade aluminium — 2026

- Reworked a 9.1 kg bronze nut into a 5.14 kg EN AW-5083 carriage, with all precision bores machined in one setup

Gear shaft and bearing housing · production drawings — 2026

- Full gear data block (ISO 1328-1 class 8), envelope requirement and run-out; housing wall raised from 1.5 to 3.5 mm

BACKGROUND

Product Development Engineer · Kromberg & Schubert R&D, Tunisia — Sep 2025 – Sep 2026

- Implemented over 100 AUDI engineering changes on 2D harness drawings and released complete packages through 6 design-freeze milestones

Final Year Project Engineer · STEG, Radès Power Plant, Tunisia — Feb 2025 – Jul 2025

- Designed an automatic seawater-basin cleaning system in SolidWorks, with an estimated reduction of over 50% in intervention time

TOOLS & LANGUAGES

CAD & simulation	SolidWorks (CSWP, CSWA, Drawing Tools), SolidWorks Simulation
Manufacturing	Sheet metal, machining, CNC programming (Mastercam, CIMCO)
Languages	Arabic (native), French (bilingual), English (B2, IELTS Academic 6.5), Italian (A1, studying daily)

EDUCATION

MSc in Mechanical Engineering · University of Cassino and Southern Lazio, Italy — Oct 2026 – Jun 2028 (expected)

Mechanical Engineering Degree (5-year programme, EUR-ACE accredited) · ENIM, Monastir, Tunisia — 2022 – 2025

Design, Numerical Simulation, Industrial Optimization

Preparatory cycle · Nabeul, Tunisia — 2020 – 2022

Availability: about 20 hours a week, Central European Time · fixed quote within 24 hours · send files to **bachraouimahdi@gmail.com**